



KubeVirt Web Management Platform

Modern, real-time web UI for managing KubeVirt virtual machines on Kubernetes — console access, monitoring, snapshots, and full host visibility.

Rust + React — Built for Production KubeVirt Clusters

Why v9s?

The management layer your KubeVirt cluster needs.



Complete VM Lifecycle

Start, stop, restart, delete, and live-migrate VMs from a single web interface. No kubectl required for day-to-day operations.



In-Browser Console

VNC graphical console with Ctrl+Alt+Delete sendkey support, serial terminal, and SSH access — all directly in the browser.



Real-Time Metrics

WebSocket-streamed CPU, memory, network I/O, and disk I/O metrics with live charts per VM. No Prometheus setup required.



Full Host Visibility

Hardware inventory (PCI, RAM modules), disk partitions, network interfaces, and systemd services — everything in one dashboard.

10

Detail Tabs per
VM

3

Console Types

5

Host Info
Sections

<50MB

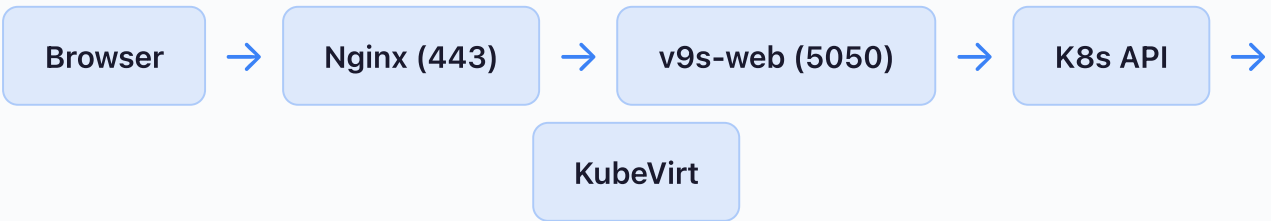
Single Binary

Technology Stack

High-performance Rust backend with a modern React frontend.

Component	Technology	Why
Backend	Rust + Axum	Async, memory-safe, single binary deployment
Frontend	React + TypeScript + Vite	Fast builds, type safety, component-based UI
State Management	TanStack Query + Zustand	Cache-first data fetching, lightweight stores
Charts	Recharts	Real-time area/line charts for metrics
VNC Console	noVNC (react-vnc)	HTML5 VNC with RFB protocol sendkey
Serial Console	xterm.js	Full terminal emulation in browser
K8s Client	kube-rs	Native Rust K8s client with watch/stream
Auth	PAM + JWT	System-level login, token-based sessions
Proxy	Nginx	TLS termination, WebSocket upgrade

Deployment Architecture



Single binary + static assets. No database. No external dependencies beyond
Kubernetes.

VM Management

Complete lifecycle control with a 10-tab detail panel.



Start / Stop / Restart

One-click lifecycle actions with KubeVirt RunStrategy API. Toast notifications for success/failure.



Delete with Auto-Close

Confirmation dialog. Details panel auto-closes after successful deletion via vmStore state management.



Live Migration

Migrate running VMs between nodes. View migration history with source/target and completion status.

VM Details Panel — 10 Tabs

Tab	Capabilities
Summary	Status, resources, tags, action buttons
Console	VNC (with sendkeys), serial terminal, SSH — console type selector
Metrics	Real-time CPU, memory, network I/O, disk I/O charts via WebSocket
Storage	Attached volumes, disk type/size, ISO mount with duplicate protection
Network	Interfaces with IP/MAC, guest OS info, add network (pod/multus, bridge/masquerade/SR-IOV)
Snapshots	Create, list, delete snapshots with ready/pending status badges
Migration	Live migrate to another node, migration history timeline

Events	Kubernetes events filtered by VM with type/reason/timestamp
Logs	Console and compute container logs from virt-launcher pod
YAML	Full VM YAML editor with copy-to-clipboard and live apply

Console Access

Three console types with full keyboard control.



VNC Graphical Console

Full desktop experience via noVNC. Scales to viewport. Fullscreen mode. Auto-reconnect on disconnect.

Sendkey Toolbar

- **Ctrl+Alt+Delete** — Login screen / task manager
- **Ctrl+Alt+F1** — Switch to text console
- **Ctrl+Alt+F2** — Switch to TTY2
- **Ctrl+Alt+F7** — Switch to graphical desktop

Uses XK keysym codes via RFB protocol. Keys that browsers intercept are sent directly to the VM.



Serial Console

xterm.js terminal emulation via KubeVirt serial console WebSocket. Ideal for headless VMs, boot troubleshooting, GRUB access.

SSH Console

Web-based SSH terminal for VMs with network connectivity. Configurable port and credentials. Direct IP-based connection.

Storage & Network

Disk management, ISO mount, and network interface control.



Storage Management

- View attached volumes with type, size, and storage class
- **Mount ISO** — Attach container disk images as CDROM (SATA)
- Duplicate protection — replaces existing CDROM entries
- Modal dialog with loading state, error display, auto-close on success
- Storage class listing for PVC provisioning



Network Management

- View interfaces with IP, MAC, state from VMI guest agent
- **Add Network** — Pod or Multus network types
- Binding models: Bridge, Masquerade (NAT), SR-IOV
- Duplicate name prevention with inline error
- Guest OS information (OS name, version, kernel)

Snapshot Management



Create

One-click snapshot creation. Auto-generated name with timestamp. Loading state



List & Filter

Filtered by VM name AND namespace. Ready/pending status badges.



Delete

Per-snapshot delete with confirmation. Loading indicator per item. Auto API

during K8s
processing.

Dedicated page +
VM details tab.

version detection
(v1beta1/v1alpha1).

Host Information

Complete hardware and system inventory in 5 tabbed sections.



Overview

Hostname, OS, kernel, architecture, CPU model, vendor, BIOS. CPU/memory/disk usage gauges with gradient bars. Uptime and process count.



Hardware

RAM Modules: Locator, size (MB), speed (MHz), type (DDR4/5), manufacturer from dmidecode. **PCI Devices:** Slot, class, vendor, device from lspci. Scrollable table.



Partitions

All filesystems: device, mount point (/boot, /boot/efi, /), type, size, used, available. Color-coded usage bars (blue/yellow/red).



Network & Services

Network: All interfaces with state, MAC, IPs, RX/TX, speed, MTU. **Services:** All systemd services with running/exited/failed badges and descriptions.

Real-Time

10-second auto-refresh

55+

System Services Tracked

15+

PCI Devices Detected

DDR4

RAM Module Details

Monitoring & Metrics

Real-time performance data via WebSocket streaming.



CPU Usage

Real-time area chart with green gradient.
Percentage display.
60-point history buffer.



Memory Usage

Blue gradient area chart. Absolute and percentage values.
Tracks allocation vs usage.



Network I/O

Dual-line chart: RX (purple) and TX (amber). KB/s throughput.
Bidirectional traffic view.



Disk I/O

Dual-line chart: Read (cyan) and Write (red). KB/s throughput.
Identifies storage bottlenecks.



Metrics Store

Zustand-based client-side store.
Per-VM history (60 data points per metric). No re-fetch on tab switch.
WebSocket auto-reconnect.

Events & Logs



Kubernetes Events

Filtered by VM. Normal/Warning type badges. Reason, message, timestamp. Tracks scheduling, lifecycle, and error events.



VM Logs

Two sources: Console log (guest output) and Compute log (QEMU/libvirt). Terminal-style

rendering. Refresh button and
source selector.

Security & Access

Enterprise authentication and authorization.



PAM Authentication

Login via Linux PAM. Supports local accounts, LDAP, SSSD, and any PAM-compatible backend. No separate user database.



JWT Sessions

Token-based sessions with configurable expiry. Bearer token on every API request. Secure cookie storage in browser.



HTTPS / TLS

Nginx TLS termination with self-signed or CA certificates. All traffic encrypted. Security headers (X-Frame-Options, nosniff).



Kubernetes RBAC

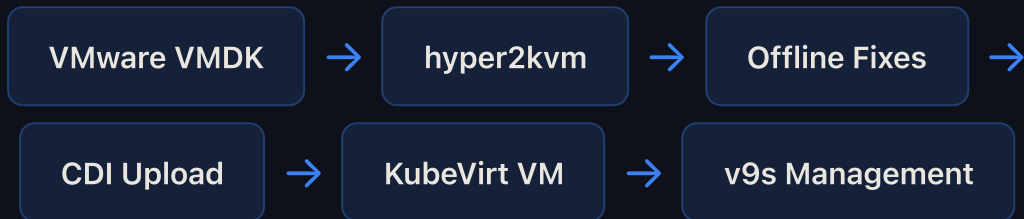
VM operations respect K8s RBAC. Users only manage VMs in authorized namespaces. 403 on unauthorized actions.

API Security

- All endpoints require Bearer token authentication
- Input validation — K8s name format enforcement on all user-supplied fields
- CORS and security headers enforced
- Request/response logging with tracing
- WebSocket connections authenticated before upgrade

hyper2kvm Integration

Complete migration-to-management pipeline.



hyper2kvm handles the migration pipeline: format conversion, bootloader repair, driver injection, and KubeVirt deployment. Once VMs are running, v9s provides the ongoing management layer — monitoring, console access, snapshots, and day-2 operations.



Migrate

hyper2kvm converts VMDK/OVA/VHD to KubeVirt with first-boot success. 8 formats, 35+ OS versions.



Deploy

Automated CDI upload, PVC creation, VM manifest. One command from VMDK to running VM.



Manage

v9s provides web console, metrics, snapshots, network, storage, and host monitoring for ongoing operations.

Ready to modernize your VM infrastructure?

From VMware to KubeVirt with hyper2kvm migration + v9s management.
Zero downtime. First-boot success. Complete visibility.

v9s — KubeVirt Web Management Platform

Part of the hyper2kvm ecosystem

github.com/ssahani/hyper2kvm